PROGRAMMING FOR PROBLEM SOLVING

Strings

What is String?

- A string is a sequence of characters terminated with null character ‘\0’.
- In C string is defined using array of characters.
- The string can be defined as the one-dimensional array of characters terminated by a null ('\0').

Example: char name [] = “COMPUTER”;

C	O	M	P	U	T	E	R	\0

- Each character in the array occupies one byte of memory, and the last character must always be 0.

What is String?

There are two ways to declare a string in c language.

- By char array
- By string literal

example: Declaring string by char array in C language.

char ch[10] = {‘c’, ’p’, ‘r’, ‘o’, ‘g’, ‘r’, ‘a’, ‘m’, ‘s', '\0’};

c	p	r	o	g	r	a	m	s	\0

char ch[] = {‘c’, ’p’, ‘r’, ‘o’, ‘g’, ‘r’, ‘a’, ‘m’, ‘s', '\0’};

without size also accepted.

What is String?

- We can also define the string by the string literal in C language.

For example: char ch[]=“cprograms";

String declaration:

Syntax : char string_name [no of characters];

example: char s1[10]; char faculty[30];

String initialization:

Syntax : char string_name [] = “”;

example: char s1[] = “c language”; char faculty[] = “Arts”;

String Example

```
#include<stdio.h>
#include <string.h>
int main(){
  char ch[10]={‘c', ‘p', ‘r', ‘o', ‘g', ‘r', ‘a', ‘m', ‘s', '\0'};
   char ch2[10]=“cprograms";

   printf("Char Array Value is: %s\n", ch);
   printf("String Literal Value is: %s\n", ch2);
 return 0;
}
```

String Example

Output

Char Array Value is: cprograms
String Literal Value is: cprograms

Traversing String

there are two ways to traverse a string.

- By using the length of string
- By using the null character.

Using the length of string

```
#include<stdio.h>
void main ()
{
    char s[10] = “cprograms";
    int i = 0;
    int count = 0;
    while(i<10)
    {
        if(s[i]=='a' || s[i] == 'e' || s[i] == 'i' || s[i] == ‘o' || s[i] == ‘u')
        {
            count ++;
        }
        i++;
    }
    printf("The number of vowels %d",count);   }
```

The number of vowels 2

Using the null character

```
#include<stdio.h>
void main ()
{
    char s[11] = “cprograms";
    int i = 0;
    int count = 0;
    while(s[i] != NULL)
    {
        if(s[i]=='a' || s[i] == 'e' || s[i] == 'i' || s[i] == ‘o' || s[i] == ‘u')
        {
            count ++;
        }
        i++;
    }
    printf("The number of vowels %d",count);   }
```

The number of vowels 2

Accepting string as the input

```
#include<stdio.h>
void main ()
{
    char s[20];
    printf("Enter the string?");
    scanf("%s",s);
    printf("You entered %s",s);
}
```

Enter the string? practice c programs

You entered practice

Accepting string as the input

It is clear from the output that, the above code will not work for space separated strings.

To make this code working for the space separated strings, the minor changed required in the scanf function,

i.e., instead of writing scanf("%s",s), we must write: scanf("%[^\n]s",s)

which instructs the compiler to store the string s while the new line (\n) is encountered.

Accepting string as the input

```c
#include<stdio.h>
void main ()
{
    char s[20];
    printf("Enter the string?");
    scanf("%[^\n]s",s);
    printf("You entered %s",s);
}
```

```
Enter the string? practice c programs
You entered practice c programs
```

String Functions

S.No	Function	Description
1)	strlen(string_name)	returns the length of string name.
2)	strcpy(destination, source)	copies the contents of source string to destination string.
3)	strcat(first_string, second_string)	concats or joins first string with second string. The result of the string is stored in first string.
4)	strcmp(first_string, second_string)	compares the first string with second string. If both strings are same, it returns 0.
5)	strrev(string)	returns reverse string.
6)	strlwr(string)	returns string characters in lowercase.
7)	strupr(string)	returns string characters in uppercase.

strlen()

The strlen() function is used to find the length of a string. It returns the number of characters in a string, excluding the null terminator (‘\0’).

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[] = "Gfg";

      // Finding and printing length of string s
    printf("%lu", strlen(s));
    return 0;
}
```

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strcpy()

The strcpy() function copies a string from the source to the destination. It copies the entire string, including the null terminator.

```
#include <stdio.h>
#include <string.h>

int main() {
    char src[] = "Hello";
    char dest[20];

    // Copies "Hello" to dest
    strcpy(dest, src);
    printf("%s", dest);
    return 0;
}
```

Hello

strncpy()

```c
#include <stdio.h>
#include <string.h>

int main() {
    char src[] = "Hello";
    char dest[20];

    // Copies "Hello" to dest
    strncpy(dest, src, 4);
    printf("%s", dest);
    return 0;
}
```

Hell